Software Requirements Specification

Automated Medical Diagnostics System

Skin Cancer Classification using Deep Learning

Based on the HAM10000 Dataset

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This document specifies the complete requirements for the development of an Automated Skin Cancer Classification System using Convolutional Neural Networks and the publicly available HAM10000 dermoscopic image dataset.

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete and unambiguous description of the requirements for the **Automated Medical Diagnostics System for Skin Cancer Classification**. The SRS translates the identified problem into measurable software, data, machine learning, interface, performance, and validation requirements. The system aims to assist dermatologists, medical students, and researchers by automatically analyzing dermoscopic images and providing a skin-lesion classification result using deep learning techniques.

This document is intended for project stakeholders including developers, project supervisors, evaluators, and future researchers who may extend the system.

1.2 Problem Statement

Skin cancer is one of the most common forms of cancer, and the visual assessment of skin lesions can be difficult because different benign and malignant lesions may exhibit similar visual characteristics. Conventional diagnosis relies heavily on expert dermatological examination and, where required, biopsy and histopathological assessment. Such processes can be time-consuming, resource-intensive, and dependent on the availability of trained specialists.

The rapid growth of dermoscopic imaging and publicly available dermatological datasets creates an opportunity to develop computer-aided diagnostic systems based on deep learning. However, a practical challenge is to transform dermoscopic images into a reliable, reproducible, and user-friendly classification system while addressing issues such as class imbalance, image variability, preprocessing consistency, model performance, and appropriate interpretation of prediction confidence.

Therefore, the problem addressed by this project is the lack of an accessible end-to-end computer-aided system that can analyze a dermoscopic skin-lesion image and provide an automated classification result using deep learning trained on the HAM10000 dataset. The proposed system will investigate whether convolutional neural networks and transfer learning can provide sufficiently accurate classification to support, rather than replace, professional clinical assessment.

The system is specifically intended as a **clinical decision-support and research prototype**. Its predictions must not be interpreted as a definitive medical diagnosis, and final diagnostic decisions remain the responsibility of qualified healthcare professionals.

1.3 Scope

The system will accept a dermoscopic image of a skin lesion as input, perform the same preprocessing pipeline used during model development, and produce a classification result with an associated confidence score. The implementation will support the selected HAM10000 classification strategy and provide an end-to-end workflow covering data preparation, model training, evaluation, model loading, and inference through a user-friendly interface. The system is designed for research and clinical decision-support purposes only and is **not** intended to replace professional medical diagnosis.

The project uses a single publicly available dataset — the **HAM10000** dataset — containing 10,015 dermoscopic images.

1.4 Definitions, Acronyms, and Abbreviations

- **SRS** — Software Requirements Specification
- **CNN** — Convolutional Neural Network
- **HAM10000** — Human Against Machine with 10000 training images
- **ISIC** — International Skin Imaging Collaboration
- **AUC** — Area Under the Receiver Operating Characteristic Curve
- **GUI** — Graphical User Interface
- **API** — Application Programming Interface

1.5 References

1. Tschandl, P., Rosendahl, C. & Kittler, H. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data* 5, 180161 (2018).
2. ISIC Archive: https://www.isic-archive.com/
3. IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.

1.6 Overview

The remainder of this document is organized as follows:

- Section 2 provides the overall description of the system and explains how the proposed solution addresses the identified problem.

- Section 3 lists the specific functional and non-functional requirements.
- Section 4 describes system architecture and design constraints.
- Section 5 covers data requirements.
- Section 6 presents the three project objectives.
- Section 7 outlines verification and validation criteria.

2 Overall Description

2.1 Product Perspective

The Automated Medical Diagnostics System is a standalone desktop/web application that uses a trained deep learning model to classify skin lesion images. It does not depend on any external real-time medical database. All training and inference are performed using the HAM10000 dataset and a locally or cloud-hosted model.

2.2 Product Functions

The major functions of the system are:

1. Accept a dermoscopic image as input (via file upload or camera).
2. Preprocess the image (resizing, normalization, optional augmentation).
3. Classify the image into one of the diagnostic categories or into binary Benign / Malignant.
4. Display the prediction result along with confidence percentage.
5. Optionally show class probabilities for all seven original classes.
6. Maintain a simple history of previous classifications (optional).

2.3 User Characteristics

- **Primary Users:** Medical students, dermatology residents, and researchers.
- **Secondary Users:** Software developers and machine learning engineers extending the system.
- Users are expected to have basic computer literacy. Medical knowledge is helpful but not mandatory for operating the system.

2.4 Constraints

- The system must use only the HAM10000 dataset for training and evaluation.
- The model must be implementable using open-source frameworks (PyTorch or TensorFlow/Keras).
- The final deliverable must run on a standard laptop/PC with moderate GPU support (or CPU-only mode).
- No patient-identifiable information is used or stored.

2.5 Assumptions and Dependencies

- Users have access to dermoscopic or high-quality clinical images of skin lesions.
- A GPU is preferred for training but not mandatory for inference.
- The HAM10000 dataset remains publicly available under its original academic license.

3 Specific Requirements

3.1 Functional Requirements

3.1.1 FR-1: Image Input

The system shall allow the user to upload a single dermoscopic image in common formats (JPG, JPEG, PNG).

3.1.2 FR-2: Image Preprocessing

The system shall automatically resize the input image to the required model input size (e.g., 224×224), normalize pixel values, and apply any necessary preprocessing pipeline used during training.

3.1.3 FR-3: Classification

The system shall classify the input image into either:

- One of the seven original HAM10000 classes, or
- Binary classes: **Benign** or **Malignant**.

3.1.4 FR-4: Confidence Score

The system shall display the prediction confidence (probability) as a percentage.

3.1.5 FR-5: Result Visualization

The system shall present the classification result in a clear visual format (text + optional probability bar chart).

3.1.6 FR-6: Model Loading

The system shall load a pre-trained model at startup or on demand.

3.2 Non-Functional Requirements

3.2.1 NFR-1: Performance

- Inference time for a single image shall be less than 3 seconds on a standard CPU and less than 1 second on a GPU.
- The model shall achieve a minimum test accuracy of 85% (multi-class) or 90% (binary classification) on a held-out test set from HAM10000.

3.2.2 NFR-2: Usability

The user interface shall be simple and intuitive so that a non-technical medical user can operate it with minimal training.

3.2.3 NFR-3: Reliability

The system shall handle invalid or corrupted image files gracefully and display appropriate error messages.

3.2.4 NFR-4: Portability

The trained model and application shall be runnable on Windows, Linux, and macOS.

3.2.5 NFR-5: Maintainability

The source code shall be modular, well-documented, and follow standard coding conventions.

4 System Architecture and Design Constraints

4.1 High-Level Architecture

The system follows a classic machine learning pipeline:

1. **Data Layer:** HAM10000 dataset (images + metadata).
2. **Preprocessing Layer:** Resizing, normalization, data augmentation (training only).
3. **Model Layer:** Convolutional Neural Network (Transfer Learning preferred — EfficientNet, ResNet50, or DenseNet).
4. **Inference Layer:** Model prediction and post-processing.
5. **Presentation Layer:** Graphical User Interface (Streamlit / Gradio / Flask + HTML or desktop GUI).

4.2 Design Constraints

- Only open-source libraries may be used (PyTorch/TensorFlow, OpenCV, scikit-learn, etc.).
- The model must be trained exclusively on the HAM10000 dataset.
- No external paid APIs or proprietary models are allowed for the core classification task.

4.3 Hardware and Software Requirements

Component	**Minimum Requirement**
Operating System	Windows 10/11, Ubuntu 20.04+, macOS 12+
Python	3.8 or higher
RAM	8 GB (16 GB recommended)
Storage	10 GB free space
GPU (optional)	NVIDIA GPU with CUDA support
Frameworks	PyTorch 2.x or TensorFlow 2.x

Table 1: Hardware and Software Requirements

5 Data Requirements

5.1 Primary Dataset

The project uses **only** the HAM10000 dataset.

Attribute	Value
Full Name	Human Against Machine with 10000 training images
Total Images	10,015
Image Type	Dermoscopic
Number of Classes	7
Resolution	Approximately 600 × 450 pixels
Source	ISIC Archive / Nature Scientific Data
License	Academic / Research use

Table 2: HAM10000 Dataset Overview

5.2 Class Distribution (Original)

- Melanocytic nevi (nv) — 6705 images
- Melanoma (mel) — 1113 images
- Benign keratosis (bkl) — 1099 images
- Basal cell carcinoma (bcc) — 514 images
- Actinic keratosis (akiec) — 327 images
- Vascular lesions (vasc) — 142 images
- Dermatofibroma (df) — 115 images

5.3 Binary Grouping (Recommended)

- **Malignant:** Melanoma, Basal cell carcinoma, Actinic keratosis
- **Benign:** Melanocytic nevi, Benign keratosis, Vascular lesions, Dermatofibroma

5.4 Data Splitting

A standard stratified split will be used:

- Training set: 70%

- Validation set: 15%
- Test set: 15%

6 Project Objectives

The project has exactly **three** primary objectives:

Objective 1: To design and implement a deep learning model (using Convolutional Neural Networks and transfer learning) capable of classifying dermoscopic images from the HAM10000 dataset into the selected lesion categories with reliable performance.

Objective 2: To develop a complete end-to-end software system that addresses the identified accessibility and workflow gap by integrating data preprocessing, model training, evaluation, model inference, and a user-friendly interface for skin lesion image classification.

Objective 3: To rigorously evaluate the performance of the developed system using standard metrics (Accuracy, Precision, Recall, F1-Score, and AUC-ROC) on a held-out test set and demonstrate its potential as a clinical decision-support tool.

7 External Interface Requirements

7.1 User Interfaces

The system shall provide a simple graphical user interface containing:

- Image upload button / drag-and-drop area
- “Classify” button
- Display area for the original image
- Display area for prediction result and confidence score
- Optional bar chart of class probabilities

7.2 Hardware Interfaces

No special hardware interfaces are required beyond standard mouse, keyboard, and display. Optional webcam support may be added for live image capture.

7.3 Software Interfaces

- Python 3.8+
- Deep learning framework (PyTorch or TensorFlow)
- Image processing library (OpenCV / Pillow)
- GUI framework (Streamlit, Gradio, or Tkinter)

7.4 Communication Interfaces

The system is designed as a standalone application. Optional future extension may include a REST API, but this is outside the current scope.

8 System Attributes

8.1 Reliability

The system shall not crash when invalid images are uploaded. Appropriate exception handling and user-friendly error messages are mandatory.

8.2 Availability

Once the model is trained and the application is launched, the system shall be available for continuous use on the local machine.

8.3 Security

No patient data is stored permanently. Uploaded images are processed in memory and discarded after prediction (unless the user explicitly saves the result).

8.4 Maintainability

The codebase shall be divided into clear modules:

- `data_loader.py`
- `preprocessing.py`
- `model.py`
- `train.py`
- `evaluate.py`
- `app.py` (GUI)

9 Verification and Validation

9.1 Verification Methods

- Unit testing of individual modules
- Integration testing of the complete pipeline
- Visual inspection of preprocessing results

9.2 Validation Criteria

The system will be considered successful if:

1. The trained model achieves $\geq 85\%$ accuracy (multi-class) or $\geq 90\%$ accuracy (binary) on the test set.
2. AUC-ROC score is ≥ 0.90 for binary classification.
3. The end-to-end application correctly classifies sample images from the test set.
4. All three project objectives are demonstrably met.

9.3 Acceptance Criteria

- Complete source code with documentation
- Trained model weights
- Performance evaluation report (confusion matrix, classification report, ROC curve)
- Working user interface
- This SRS document and a final project report

10 Appendices

10.1 Appendix A — Glossary

Dermoscopic Image A high-resolution image of a skin lesion taken with a dermatoscope.

Transfer Learning Technique of using a pre-trained model (usually on ImageNet) and fine-tuning it on a new dataset.

Data Augmentation Artificial expansion of the training set by applying transformations (rotation, flip, zoom, etc.).

10.2 Appendix B — Future Enhancements (Out of Scope)

- Multi-disease classification beyond pigmented lesions
- Mobile application version
- Integration with hospital information systems
- Explainable AI (Grad-CAM visualization)

— End of Software Requirements Specification —